

OASIS — Identified Formula Errors

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Scientific validation of the OASIS computational engine

Date: 2026-07-03 · Branch: `feat/detailed-ecosystemic-report` · Scope: all 99 inventoried formulas in `src/`

How to read this report

This report documents every computation error identified by an exhaustive validation of the OASIS codebase, in which each of the 99 scientific/mathematical formulas was cross-checked against the peer-reviewed papers in `_papers/` (Ulanowicz 2009; Zorach & Ulanowicz 2003; Fath 2019; Finn 1976; Levine 1980; and canonical network-science references) or, for the proprietary OASIS composite, against internal design logic.

Each error carries a **verification status**:

- ✔ **Confirmed** — high confidence; a logic error, a documentation/consistency contradiction, or a data-provenance problem that does not depend on a contested scientific interpretation.
- ✔ **Expert-confirmed** — a scientific/mathematical correction that a three-member adversarial expert panel (mathematician, ecosystem-dynamics theorist, ENA methodologist) independently re-derived and confirmed against the papers.
- ⚖ **Expert-reviewed — reclassified** — a claim the panel examined and **downgraded** from a “fix” to a design/science *decision* (it is not unambiguously demanded by the literature). Per the project rule “no formula change without peer-reviewed support,” these are **not** auto-implemented.

Governing rule: scientific formulas are not changed unless a peer-reviewed paper unambiguously supports the correction.

Executive summary

The core mathematics is sound. All 11 core Ulanowicz information-theoretic measures — Total System Throughput, Average Mutual Information, Ascendancy (A), Development Capacity (C), Reserve/Overhead (Φ), relative ascendancy (α), flow diversity, and the identity $C = A + \Phi$ — are **correct and paper-faithful**. Marginal sums are not swapped, zero-flow terms are handled correctly, and the two independent implementations (loop and vectorized) agree to machine precision. **No headline number produced by the core engine is mathematically wrong.**

The errors are concentrated in the layers built on top of that core — the derived metrics, the threshold constants, the proprietary 5-dimension composite, and the report/presentation layer.

27 distinct defects were identified:

Severity	Count	Nature
Critical	3	Directly corrupt a headline verdict or the sustainability score

Severity	Count	Nature
Major	14	Wrong derived metric, wrong network-science formula, or self-contradicting report
Minor	10	Documentation, labeling, and normalization-transparency issues

The single most important finding is not a formula at all — it is a **design flaw in how the five dimensions are combined**: the overall health verdict can read “HEALTHY” while the organization is “Non-Viable,” because a flat average lets three strong dimensions mask a collapsed one. This is the root cause of the credibility problem and is fixable with a small, explainable change to the roll-up logic.

The most consequential scientific finding — after adversarial expert review — is not a fix, it is a caution. An initial pass proposed globally re-targeting the “optimal α ” from 0.37 to **0.4596**, citing Ulanowicz (2009). A three-member expert panel (mathematician, ecosystem-dynamics theorist, ENA methodologist) **overturned that as a clean fix**. They established that (i) 0.4596 is an *empirically calibrated* value, not a mathematical theorem (the exponent $\beta = 1.288$ is back-solved from it, so the two are circular); (ii) the organization-facing paper, **Fath (2019), itself defines robustness as $-\alpha \log \alpha$ and maximizes it at $1/e \approx 0.37$** , so 0.37 is *not* simply wrong; and (iii) a global swap would make the robustness curve’s peak ($1/e$) and the “optimum” (0.4596) contradict each other. **Most importantly, the ecologist found that these ecological viability optima are not established to transfer to organizations at all** — Fath (2019) explicitly notes economic/organizational networks are more redundant, sit in a different region of the curve, and that their calibration is an open research question. This means the product’s “every organization is unsustainable” pattern is most likely a **mis-calibrated window artifact, not a true diagnosis** — a finding that matters more for a sellable product than any single formula. Details in §1 (E-2) and the panel verdicts below.

1. Critical errors

E-1 · The composite roll-up has no viability floor Confirmed

Where: `oasis_calculator.py:695–698, 713–718` · **Classification:** proprietary-design decision

What happens. The overall OASIS health score is a flat weighted average of the five dimensions (20% each), and the overall HEALTHY/WARNING/CRITICAL band is applied to that average *independently* of any single dimension’s status. So an organization scoring (OPEN 100, AUTONOMOUS 100, SYMBIOTIC 100, INTELLIGENT 100, SUSTAINABLE 0) averages to **80** → “HEALTHY” even though its SUSTAINABLE dimension is CRITICAL and the organization is Non-Viable. There is no floor, veto, or worst-dimension rule anywhere.

Why it matters. This is the direct cause of the product’s most damaging contradiction — a report that simultaneously says “Non-Viable” and “76/100 HEALTHY.” It is amplified by the normalization caps (see E-24), which pin the other three dimensions near 100 and let them outvote a collapsed SUSTAINABLE.

Fix (design decision required). Add a viability veto: the overall status cannot be HEALTHY if any dimension — especially SUSTAINABLE / Window-of-Viability — is CRITICAL. This is the smallest, most client-explainable change; it corrects the headline verdict without altering any underlying number. Alternative roll-ups (geometric/harmonic mean, multiplicative sustainability gate) are stronger but require re-baselining and are deferred as product choices.

E-2 · The “optimal α ” target and the organizational-calibration question Expert-reviewed — reclassified

Where: `oasis_calculator.py:623–626` (α -optimality) · **Classification:** proprietary/scientific design decision (NOT a clean paper-backed fix)

What happens. The α -optimality sub-score rewards proximity to $\alpha = 0.37$ ($\approx 1/e$), the peak of the robustness proxy $R = -\alpha \ln \alpha$. An initial validation proposed re-targeting this to **0.4596** (Ulanowicz 2009’s “window of vitality” center) as a paper-backed correction.

Expert-panel verdict (adversarial review). The panel **partially refuted** that proposal:

- **Mathematician:** $0.4596 = e^{-(1/\beta)}$ with $\beta = 1.288$, but β is itself *back-solved* from the 0.4596 window center — the two are circular by construction. So 0.4596 is an *empirical calibration*, not a mathematically forced optimum. The paper hedges it as provisional.
- **Ecosystem-dynamics theorist:** the claim that theory “rejects $1/e$ ” is **refuted** — the same author’s *Dual Nature* paper (2009) calls $\alpha = 1/e$ “the point of natural sustainability,” and Fath (2019) — the paper that extends this to economics/organizations — defines “Systemic Robustness = $-\alpha \log \alpha$ ” and maximizes it, i.e. at $1/e$. So 0.37 is a defensible target, especially for organizations.
- **Internal-consistency catch:** if the code keeps $R = -\alpha \ln \alpha$ (which peaks at $1/e$) but moves the *target* to 0.4596, the robustness peak and the “optimum” contradict each other — they come from two different kernels.

The deeper finding (why this matters most). The bigger issue is not 0.37-vs-0.4596 but **whether an ecological viability window applies to organizations at all**. Fath (2019) explicitly states economic and organizational networks are *more redundant, less efficient, and sit in a different region of the curve* than ecosystems, and that their calibration is an **open research question**. The observed pattern — every organization sample ($\alpha \approx 0.07\text{--}0.10$) reads “unsustainable,” only a literal wetland ($\alpha \approx 0.58$) passes — is therefore most consistent with a **mis-transferred window / scale artifact, not genuine organizational dysfunction**.

Recommendation. Do **not** perform a global $0.37 \rightarrow 0.4596$ swap (scientifically indefensible and self-contradicting). Instead treat the organizational optimum and window as a **calibration decision**: keep $1/e$ where it legitimately normalizes the robustness proxy (the $R/(1/e)$ normalization, which the mathematician confirmed is correct), keep the ecosystem-only $\beta = 1.288$ path where it already correctly uses 0.4596, and **re-derive or explicitly caveat the organizational viability window** rather than inheriting the ecological one. This is the highest-value scientific decision for making the product credible.

E-3 · Regenerative-capacity center 0.37 + α /efficiency variable naming ⚖️ Expert-reviewed

Where: `ulanowicz_calculator.py:877–887` · **Classification:** tied to the E-2 calibration decision

What happens. Regenerative capacity is $R \cdot (1 - |\alpha - 0.37|)$; the α value arrives via a function named “network efficiency” (the value *is* $\alpha = A/C$, but the name misleads).

Verdict. The `0.37` here is subject to the **same E-2 calibration decision** — it is **not** an independent paper-backed fix (the panel refuted the blanket 0.4596 swap). The variable-naming confusion (efficiency vs α) is a genuine, separable clarity fix. The $R \cdot (1 - |\Delta|)$ blend shape is proprietary.

2. Major errors

Derived ecological metrics (Ulanowicz / ENA methods) ✅ Expert-confirmed

Verified by the ENA methodologist and the mathematician against Finn (1976), Levine (1980), and Zorach & Ulanowicz (2003) — all confirmed, one strengthened.

ID	Metric	Where	What’s wrong	Correct form (citation)
E-7	Effective connectivity inverted	<code>ulanowicz_calculator.py:1084–1086</code>	Computes N/F (< 1) — the reciprocal of connectivity — and violates the hard “connectivity ≥ 1 ” floor. Root cause (found by the panel): a dropped negative sign in the exponent. Confirmed	Connectivity C = F/N (flows per node, ≥ 1) — Zorach & Ulanowicz 2003, p.72/76

ID	Metric	Where	What's wrong	Correct form (citation)
			numerically: code = 0.31 where F/N = 3.22.	
E-8	"Finn Cycling Index" counts only short cycles	ulanowicz_calculator.py:719–729	Counts self-loops + 2-cycles only; confirmed to return 0.0 for a pure 4-node ring whose true cycling → 100%.	Relabel as short-cycle proxy; use the full Finn index (E-9)
E-9	Full Finn Cycling Index mis-normalized	ecosystem_flow_calculator.py:140–144	Normalizes by the scalar total instead of column throughflow, and sums the wrong Leontief entries → a systematic underestimate of cycling (panel measured 0.2–1.5× the canonical value across test networks — <i>not</i> the clean "2×" an earlier pass claimed, but wrong in direction and magnitude).	FCI = TSTc/TST via column-stochastic Leontief inverse — Finn 1976; Ulanowicz 2004 §5
E-10	Trophic depth uses topological hops	ulanowicz_calculator.py:628	Uses unweighted average shortest-path length, ignoring flow magnitudes; cannot produce fractional effective trophic levels.	Flow-weighted effective trophic level from the structure matrix — Levine 1980; Ulanowicz 2004 §4
E-11	"Lindeman efficiency" mislabeled	ecosystem_flow_calculator.py:194–196	1 – respiration/(TST+imports) is a system-wide retention ratio, not Lindeman between-level (≈10%) transfer efficiency.	Transfer efficiency from the Lindeman spine — Lindeman 1942; or rename the metric

Network-science metrics on directed flow graphs Expert-confirmed

Verified by the mathematician (sympy/numpy re-derivations) against canonical references. All confirmed; the Gini coefficient was checked and found **correct** (equals the mean-absolute-difference Gini to machine precision — no change needed).

ID	Metric	Where	What's wrong	Correct form (citation)
E-13	Betweenness/closeness treat flow as distance	network_analyzer.py:86,103	weight='weight' makes shortest paths <i>minimize</i> flow, so strong high-flow ties are treated as long/far — inverted. Betweenness feeds the OPEN dimension , so this mis-scores OPEN.	Invert to cost $d = 1/\text{flow}$ — Brandes 2001
E-14	Small-world random baseline corrupted	network_analyzer.py:230–231	Mean degree is read from the wrong function (returns average-neighbour-degree of degree-1 nodes), corrupting the random path-length baseline → σ , ω , and the small-world verdict are unreliable.	$\langle k \rangle = 2m/n$ — Fronczak et al. 2004
E-12	Freeman centralization normalizer	ulanowicz_calculator.py:956–963	Uses the undirected star maximum $(n-1)(n-2)$ on directed degrees; the value can exceed 1.	Directed normalizer $(n-1)^2$ — Freeman 1979
E-15	Small-world ω uses wrong clustering baseline	network_analyzer.py:244	Uses random-graph clustering where the ω coefficient requires <i>lattice</i> clustering.	$\omega = L_{\text{rand}}/L - C/C_{\text{lattice}}$ — Telford / Bassett 2011

ID	Metric	Where	What's wrong	Correct form (citation)
E-16	Rich-club coefficient unnormalized	network_analyzer.py:314–320	normalized=False yields a monotone, uninterpretable curve; degree cutoff is arbitrary.	normalized=True (ratio to degree-preserving randomization) — Colizza 2006
E-18	Flow-diversity utilization mixes log bases	publication_report.py:266–267	Divides a natural-log (nats) diversity by a base-2 (bits) denominator → understates utilization by ~31%.	Match bases (use $\log(n^2)$ in nats)

Report/presentation contradictions Confirmed

Code is correct; only the rendered text/bands are wrong or inconsistent.

ID	Where	What's wrong
E-19	publication_report.py , report_intelligence.py , main.py , pdf_generator.py	The same α is called “Very High / good efficiency” in one section and “over-constrained / brittle, HIGH risk” in another; the breakpoints also differ across files (0.2/0.4/0.6 vs 0.2/0.35/0.45/0.6).
E-20	publication_report.py , latex_report_generator.py , main.py	The robustness “high” threshold is 0.20 on the PDF path but 0.25 on the LaTeX/CLI path , so $R = 0.22$ flips verdict depending on which export you run.
E-21	publication_report.py Appendix	The methodology appendix prints “ Network Efficiency = $A/(C \cdot \log_2 n)$ ”, but the engine computes $A/C = \alpha$. The printed formula contradicts the number shown. (Engine is correct; fix the text.)

Benchmark data provenance Confirmed (needs source-tracing)

ID	Where	What's wrong
E-22	services/published_metrics_db.py:179–186	The stored Florida Bay $\alpha = 0.367$ cannot be sourced: the cited Heymans 2002 paper is about Everglades graminoid/cypress ecosystems (reporting $\approx 0.52 / 0.34$, never 0.367), the “seagrass/marine” label mismatches, and 0.367 suspiciously equals $1/e$ used elsewhere. A benchmark anchor should not ship with an unverifiable value.

3. Minor errors Confirmed

ID	Where	What's wrong	Note
E-23	oasis_calculator.py:599–600 vs 633–638	SUSTAINABLE docstring weights (0.30/0.25/0.20/0.25) differ from the executed weights (0.30/0.20/0.20/0.30). Both sum to 1.0 — scoring unaffected, but auditors reading the docstring get the wrong model.	Doc fix
E-24	oasis_calculator.py (per-dimension caps 0.5–0.8)	The normalization caps are necessary size-scaling devices (many metrics grow with network size, so caps make a 0–100 score comparable across a 5-node org and a 40-node ecosystem) — <i>not</i> arbitrary bugs. But they are undocumented and <i>fixed</i> , causing saturation that amplifies E-1.	Size-relative refinement + documentation
E-25	oasis_calculator.py (divisors roles/10, rolesPerNode/2, regen/0.3)	Same principle: these gauge size, but a fixed divisor assumes a size and mis-gauges very small/large networks.	Make size-relative (relative to n)

ID	Where	What's wrong	Note
E-26	ulanowicz_calculator.py:815–818	Autocatalytic index uses a $\cdot 10$ amplifier and an $n(n-1)/2$ normalizer with no theoretical basis; any network with >10% cycle flow saturates.	Proprietary blend; report raw components
E-27	precompute_pipeline.py:117–118	Two “density” definitions coexist (m/n^2 and $m/(n(n-1))$).	Pick one denominator

Additional minor items (Katz centrality parameter not adaptive, low simulation counts, direct-only mutualism, base-convention labeling) are documented in the underlying validation files.

4. Two tracks for remediation

The errors split cleanly into two remediation tracks.

Track 1 — Paper-backed / canonical corrections (expert-confirmed, safe to implement)

Each is unambiguously specified by a peer-reviewed paper or canonical reference **and confirmed by the expert panel** — permissible under the “no formula change without peer-reviewed support” rule:

- Effective connectivity = F/N (E-7) — Zorach & Ulanowicz 2003 ✓
- Full Finn Cycling Index; relabel the short-cycle proxy (E-8/E-9) — Finn 1976; Ulanowicz 2004 ✓
- Flow-weighted trophic depth (E-10) — Levine 1980 ✓
- Lindeman efficiency (relabel or replace) (E-11) — Lindeman 1942 ✓
- Betweenness/closeness distance inversion (E-13) — Brandes 2001 ✓
- Freeman $(n-1)^2$, small-world $\langle k \rangle = 2m/n$, ω lattice clustering, rich-club normalization, log-base consistency (E-12, E-14, E-15, E-16, E-18) — canonical network science ✓
- Mutualism should include *indirect* (integral-utility) effects, not direct-only — Fath 2019 Principle 8 ✓
- Report/label consistency: efficiency labels, robustness threshold, appendix formula, docstring weights (E-19, E-20, E-21, E-23) — ✓ code already correct, fix the text

Track 2 — Scientific & proprietary design decisions (require a product/science call, not a literature fix)

- **The organizational α -optimum and viability window (E-2/E-3)** — the panel refuted a blanket 0.37 $\rightarrow$ 0.4596 swap; decide the organizational calibration (keep 1/e for the robustness proxy; re-derive or caveat the org window). **This is now a design decision, not a Track-1 fix.**
- The roll-up viability veto (E-1) — which rule to adopt
- Size-relative redesign of the normalization caps and divisors (E-24, E-25) — the basis to use
- Whether to keep the [0.2, 0.6] α -window heuristic or move to the paper’s (c, n) formulation — the panel found [0.2, 0.6] is **not** in the primary literature and, applied to organizations, is what “manufactures” the near-universal fail
- The autocatalysis blend and general magic-number tuning (E-26 and the overall/threshold bands)

5. What changes if these are fixed (regression note)

Fixes that **change computed numbers** (and therefore require re-baselining and test updates): the SUSTAINABLE dimension (via the α -optimum), OPEN (via betweenness), AUTONOMOUS (via cycling and connectivity), the overall verdict (via the veto), the Finn Cycling Index ($\approx$ doubles), effective connectivity (inverts), small-world coefficients, Freeman centralization, trophic depth, rich-club, the flow-diversity utilization %, and the Florida Bay benchmark anchor.

Fixes that **change only rendered text** (no number moves): the efficiency labels, the robustness threshold wording, the appendix formula, and the docstring weights.

The published-value validation suite (`services/`) and any unit tests over `oasis_calculator` , `ulanowicz_calculator` , `vectorized_metrics` , `ecosystem_flow_calculator` , and `network_analyzer` must be re-run, and loop-vs-vectorized parity re-confirmed after the connectivity fix.

Verification status & panel outcome

The adversarial expert panel — a **mathematician**, an **ecosystem-dynamics theorist**, and an **ENA methodologist** — has **completed** its review. Each independently re-derived results and re-read the papers to *refute* rather than confirm. Outcome:

- **Confirmed (safe to implement, Track 1):** the ENA-method corrections (effective connectivity = F/N , full Finn Cycling Index, flow-weighted trophic depth, Lindeman relabel), all directed-network-science fixes (Freeman, betweenness inversion, small-world baseline, ω , rich-club, log base), and the mutualism indirect-utility extension. The Gini coefficient was checked and is **correct** (no change).
- **Confirmed (Track 1, code correct — text only):** the report/label contradictions (E-19–E-21, E-23).
- **Refuted / reclassified (now Track 2 — a decision, not a fix):** the blanket α -optimum swap to 0.4596 (E-2/E-3). The panel showed 0.4596 is an empirical calibration (not a theorem), that $1/e$ is a defensible target for organizations (Fath 2019 uses $-\alpha \cdot \log \alpha$), and that a global swap would be internally self-contradicting.
- **Elevated finding:** the ecological viability window ($[0.2, 0.6]$, optimum) is **not established to apply to organizations**; the “every organization is unsustainable” result is most likely a calibration artifact. This is the top scientific item to resolve for product credibility.

Bottom line: the engine’s core mathematics is sound; the confirmed Track-1 corrections can proceed under the peer-reviewed-support rule; and the two highest-value items — the roll-up viability veto (E-1) and the organizational calibration of the viability window (E-2) — are **product/science decisions** that should be made deliberately, not auto-fixed. The adversarial review paid off precisely by stopping a plausible-but-wrong “correction” from being applied.

Full per-formula validation detail is in `docs/business-revision/evidence/validation-*.md` , the consolidated `validation-SYNTHESIS.md` , and the three panel reports `evidence/expert-*.md` . This report documents identified errors only; no source code has been modified.